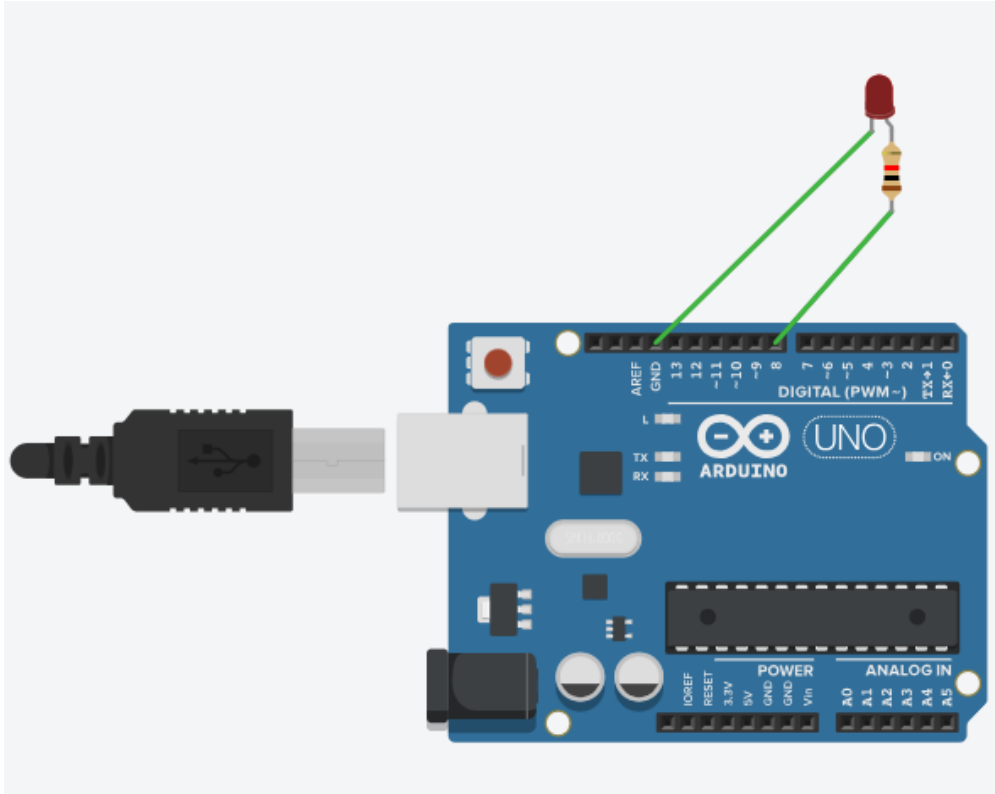


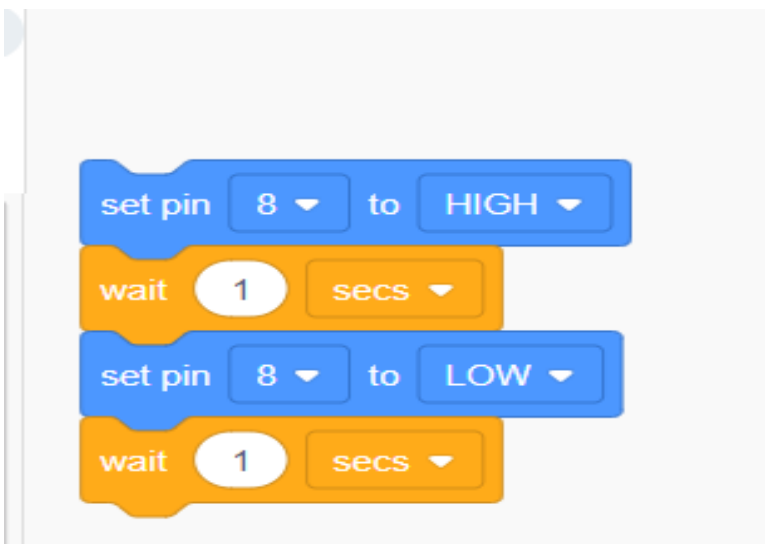
IoT projects:

1. Simple LED on and Off:

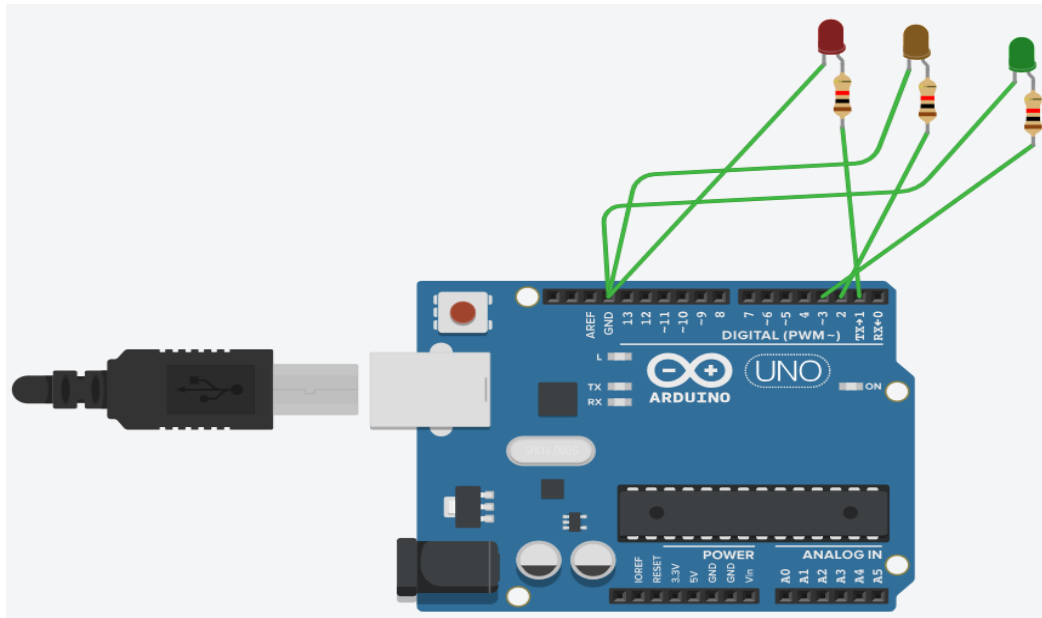
Circuit Configuration:



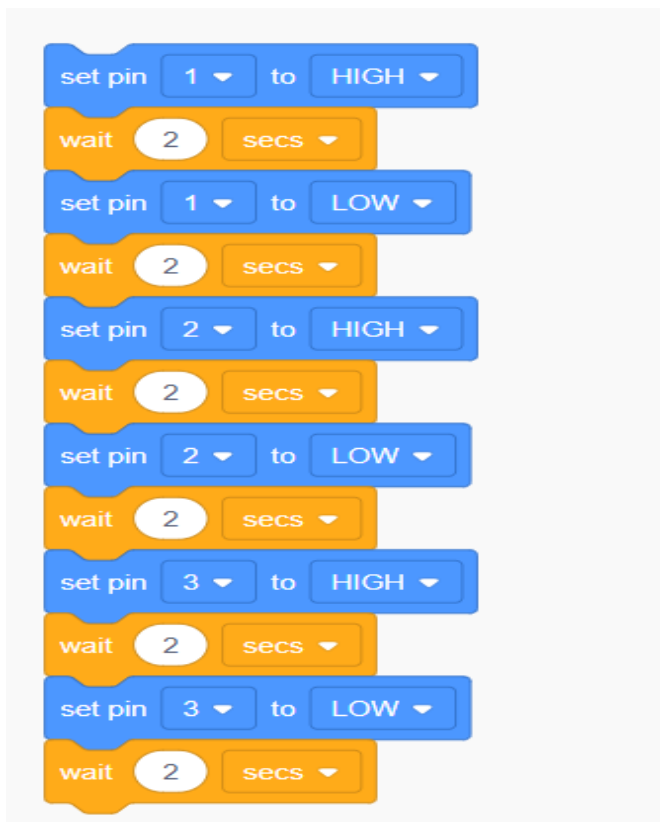
Code:



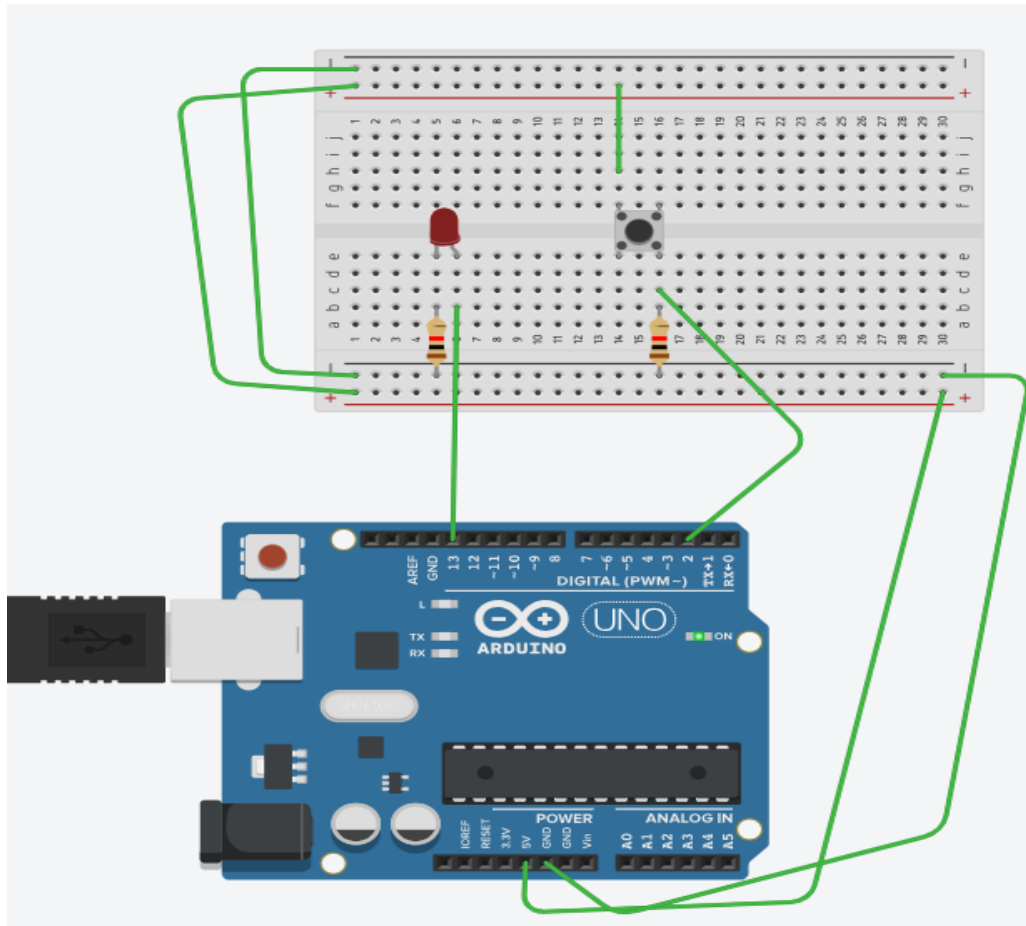
2. 3 Led RGB with a delay of 2sec:



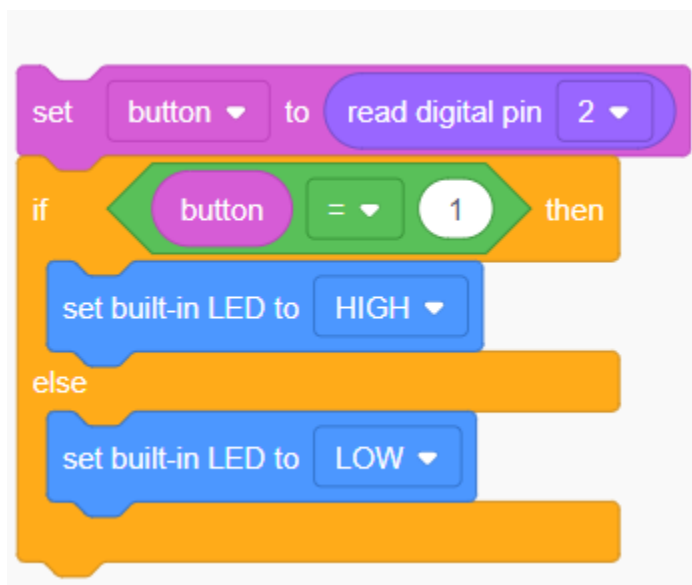
Code:



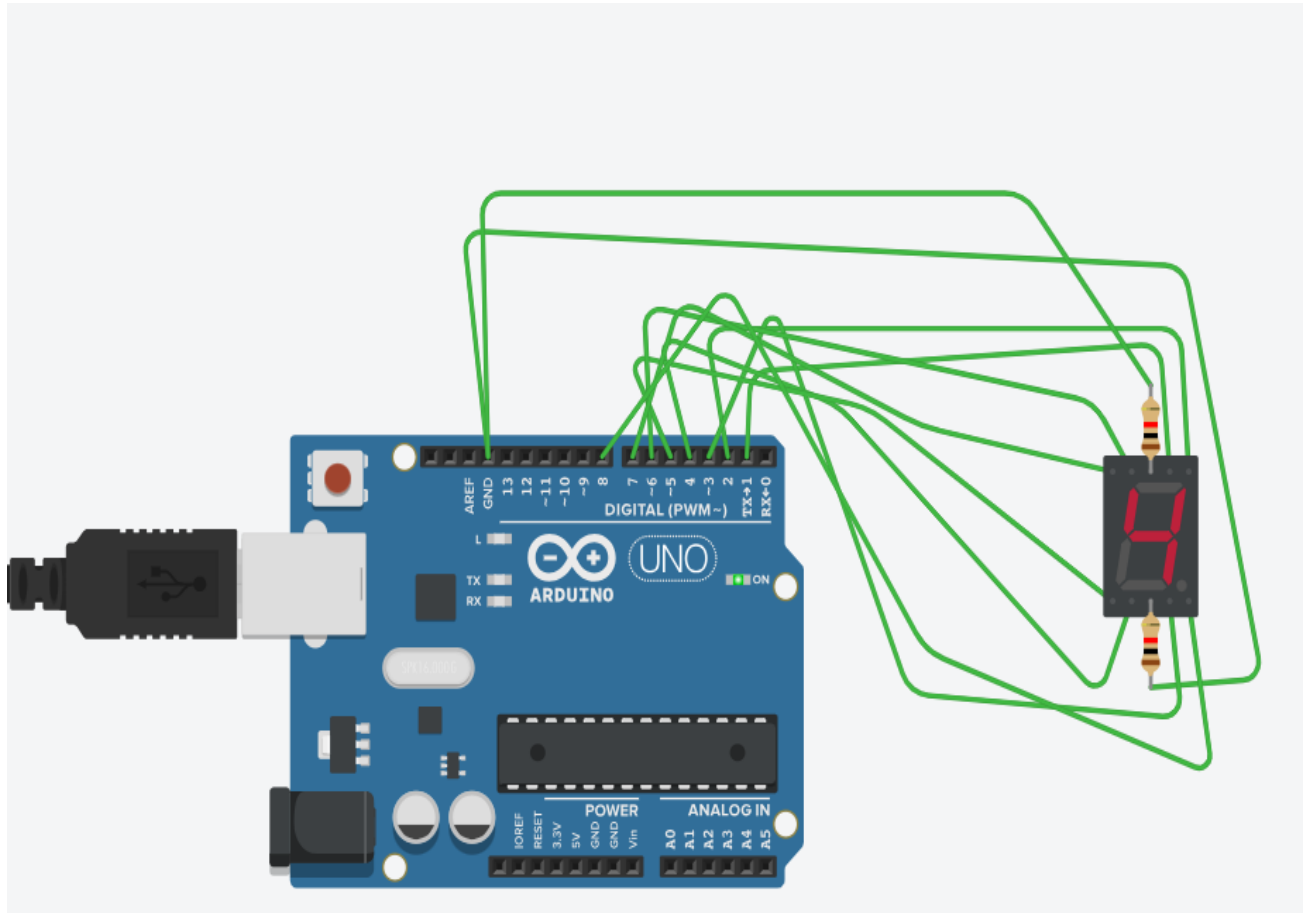
3. Led On and Off with push button:



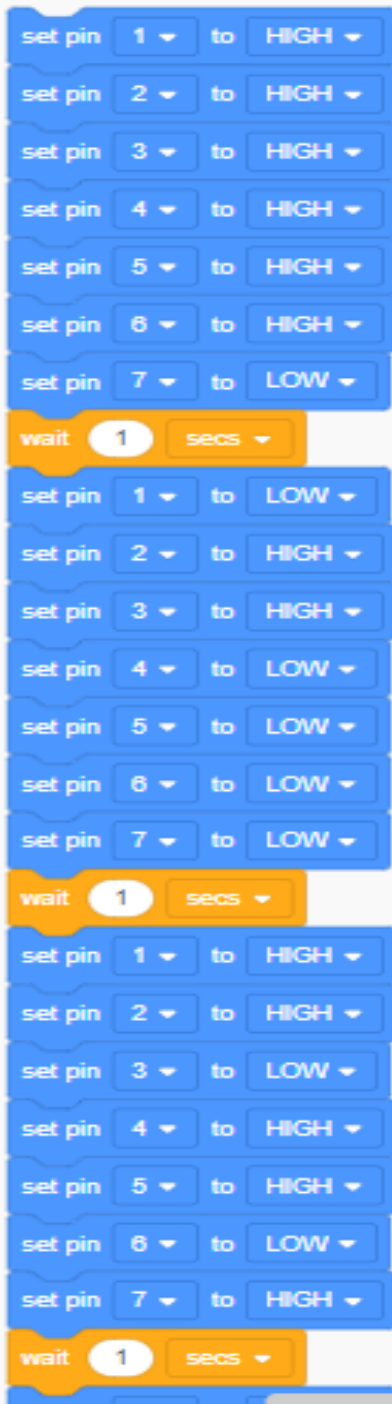
Code:



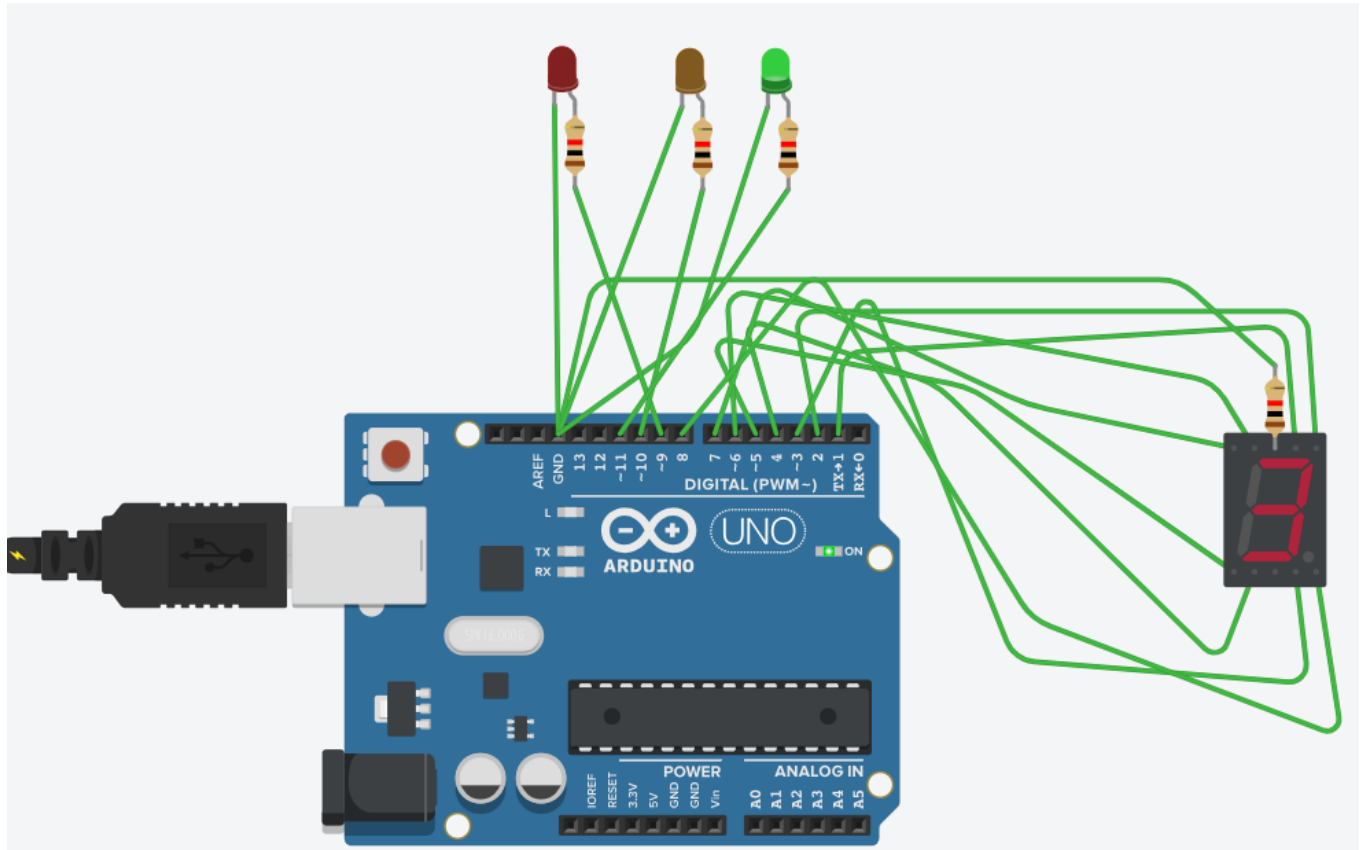
4. 7-Segment Display:



Code:



5. Seven-segment display with LED traffic light:



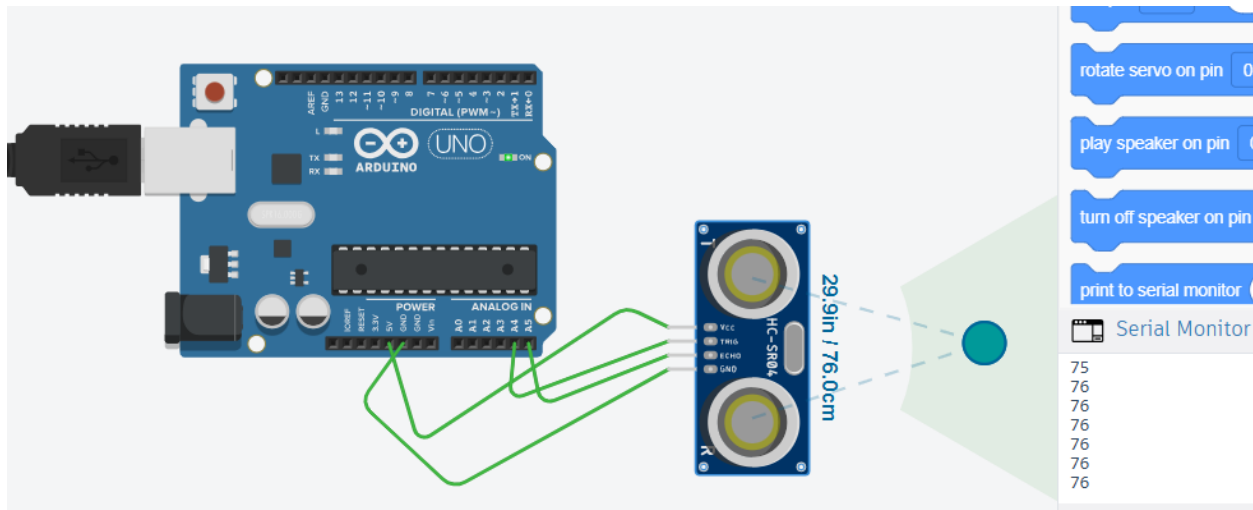
Code:

```

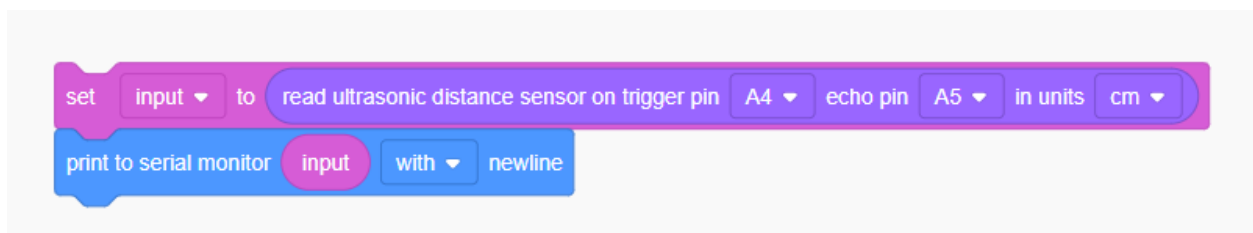
1  const int redLED      = 9;
2  const int yellowLED = 10;
3  const int greenLED   = 11;
4  const int segPins[7] = {1, 2, 3, 4, 5, 6, 7};
5  const byte digits[10][7] = {
6      {1,1,1,1,1,1,0}, // 0
7      {0,1,1,0,0,0,0}, // 1
8      {1,1,0,1,1,0,1}, // 2
9      {1,1,1,1,0,0,1}, // 3
10     {0,1,1,0,0,1,1}, // 4
11     {1,0,1,1,0,1,1}, // 5
12     {1,0,1,1,1,1,1}, // 6
13     {1,1,1,0,0,0,0}, // 7
14     {1,1,1,1,1,1,1}, // 8
15     {1,1,1,1,0,1,1}  // 9
16 };
17 void displayDigit(int num) {
18     for (int i = 0; i < 7; i++) {
19         digitalWrite(segPins[i], digits[num][i]);
20     }
21 }
22 void setup() {
23     pinMode(redLED, OUTPUT);
24     pinMode(yellowLED, OUTPUT);
25     pinMode(greenLED, OUTPUT);
26
27     for (int i = 0; i < 7; i++) {
28         pinMode(segPins[i], OUTPUT);
29     }
30 }
31
32 void loop() {
33     // GREEN light countdown (5 → 0)
34     digitalWrite(greenLED, HIGH);
35     for (int i = 5; i >= 0; i--) {
36         displayDigit(i);
37         delay(1000);
38     }
39     delay(1000);
40     digitalWrite(greenLED, LOW);
41
42     // YELLOW light countdown (2 → 0)
43     digitalWrite(yellowLED, HIGH);
44     for (int i = 2; i >= 0; i--) {
45         displayDigit(i);
46         delay(1000);
47     }
48     digitalWrite(yellowLED, LOW);
49
50     // RED light countdown (5 → 0)
51     digitalWrite(redLED, HIGH);
52     for (int i = 5; i >= 0; i--) {
53         displayDigit(i);
54         delay(1000);
55     }
56     digitalWrite(redLED, LOW);
57 }

```

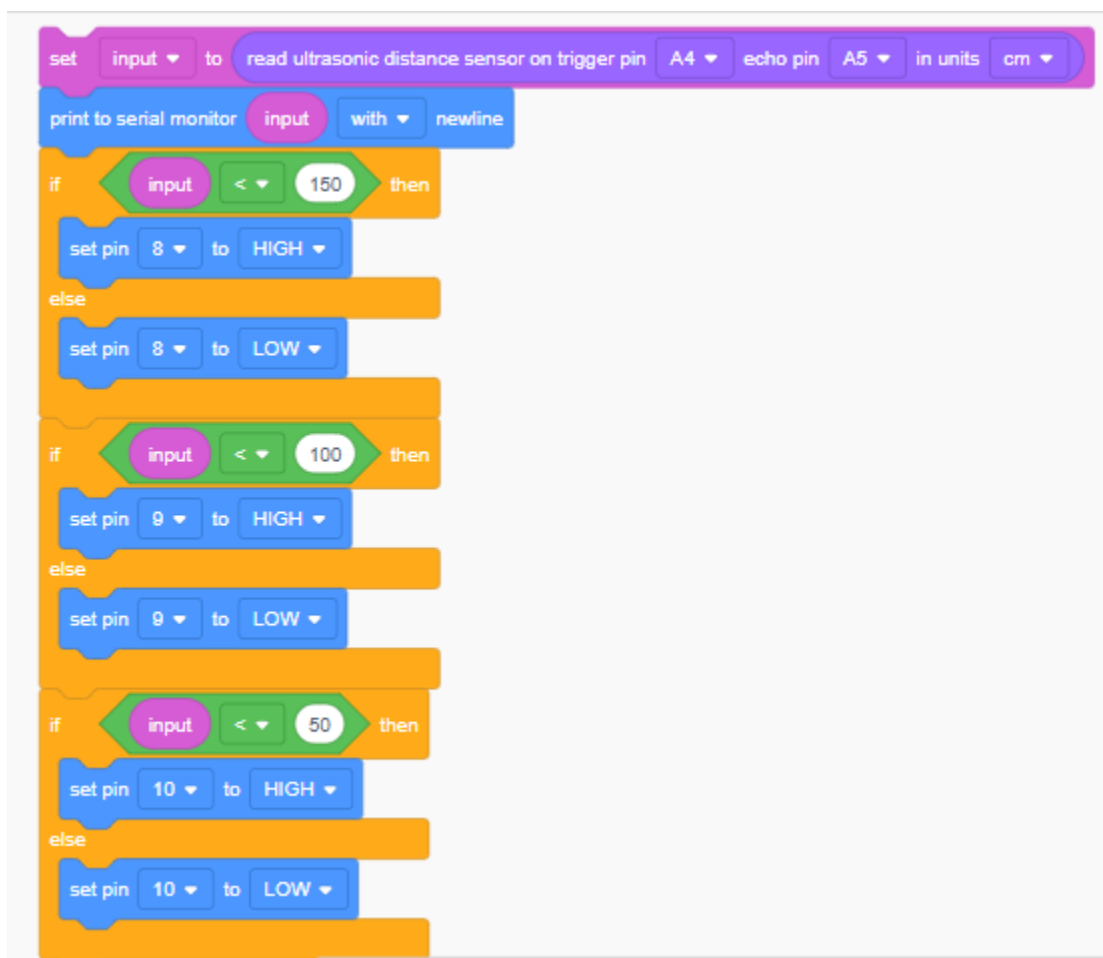
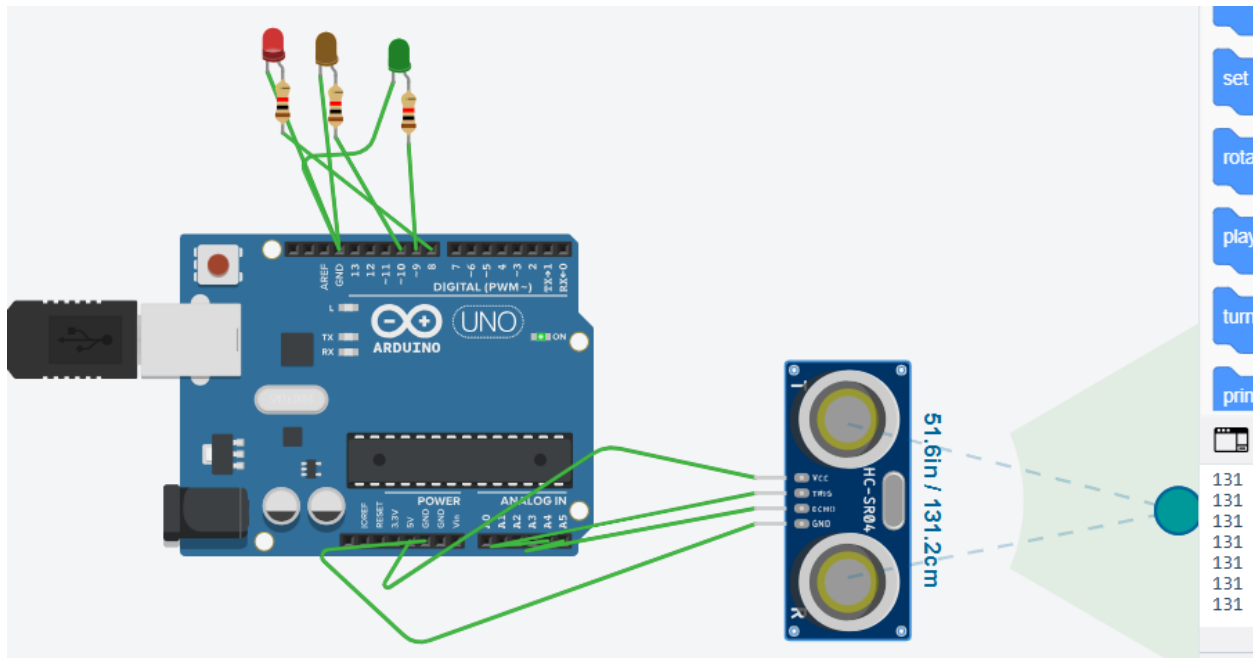
6. Ultrasonic Sensor Interfacing:



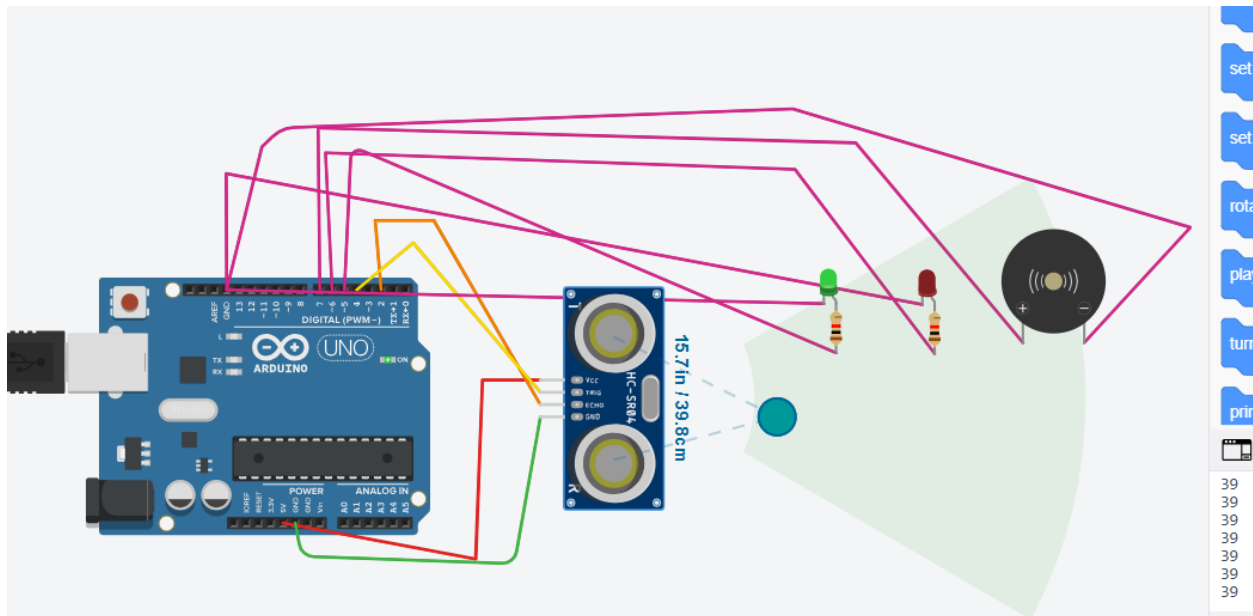
Code:



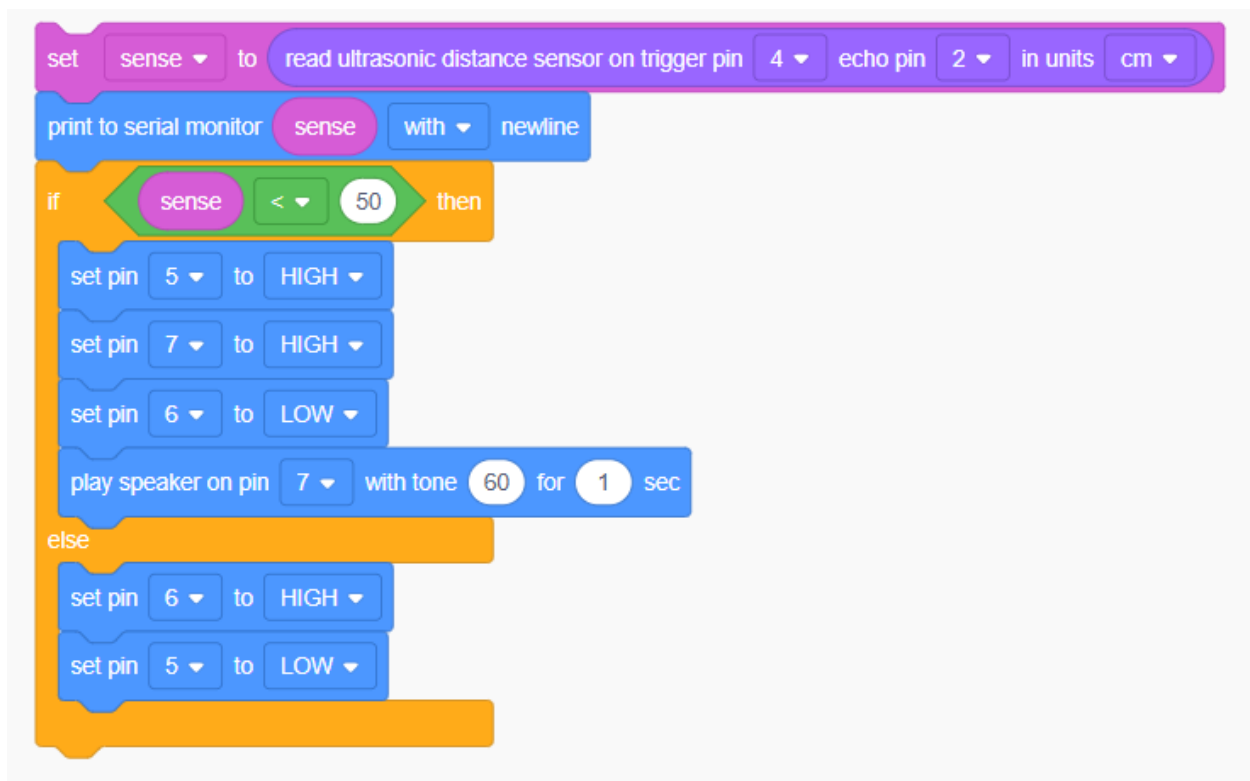
7. Distance Measurement with LED & Arduino and Ultrasonic sensors:



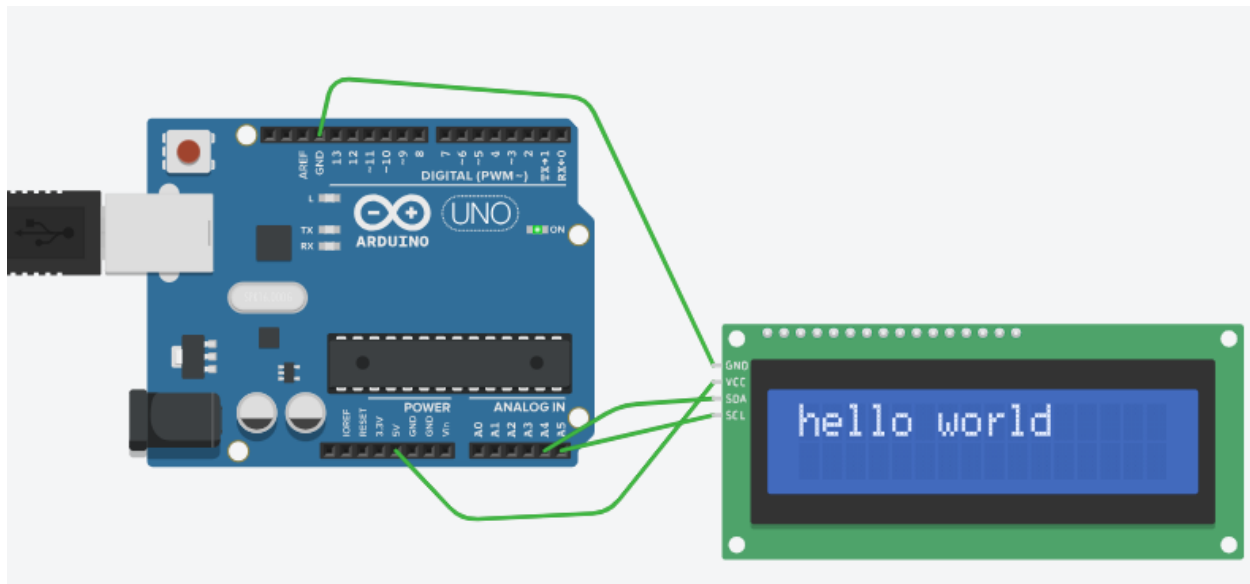
8. Distance Measurement with LED & Arduino and Ultrasonic sensors with an alarm as well:



Code:



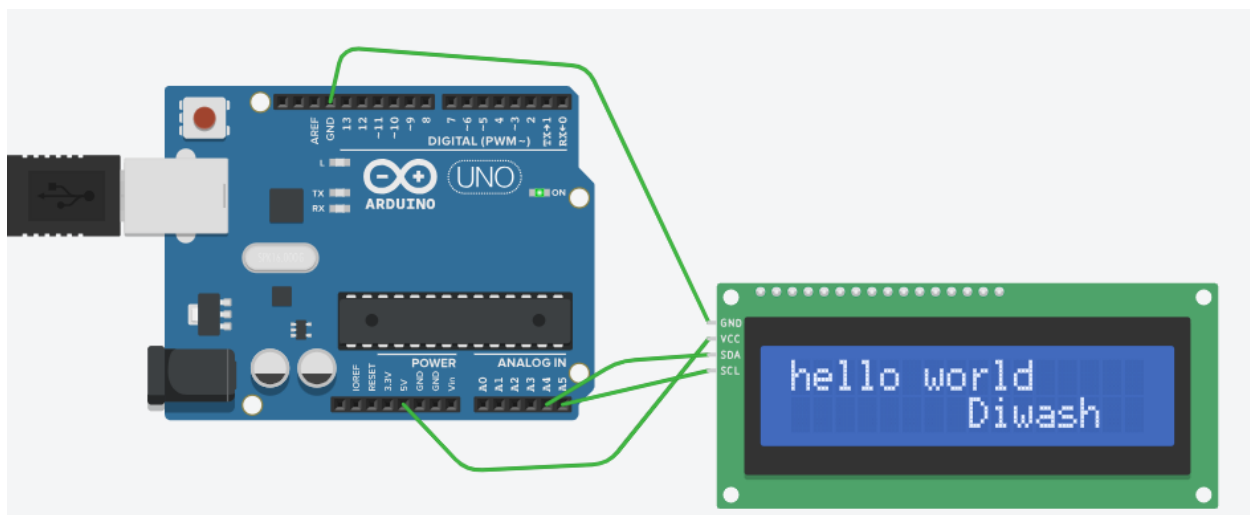
9. Print Hello world



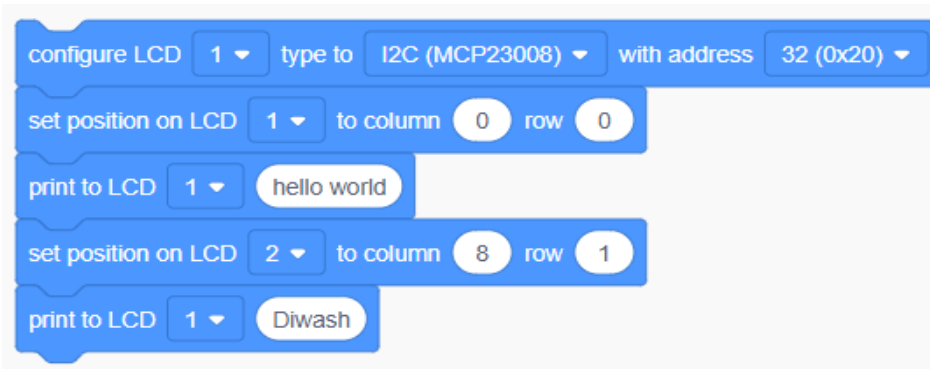
Code:



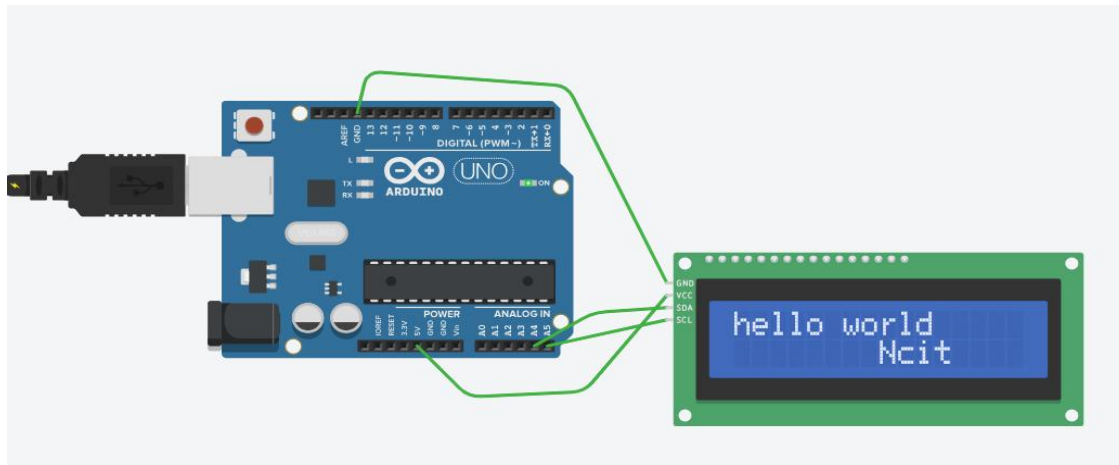
Print name:



Code:



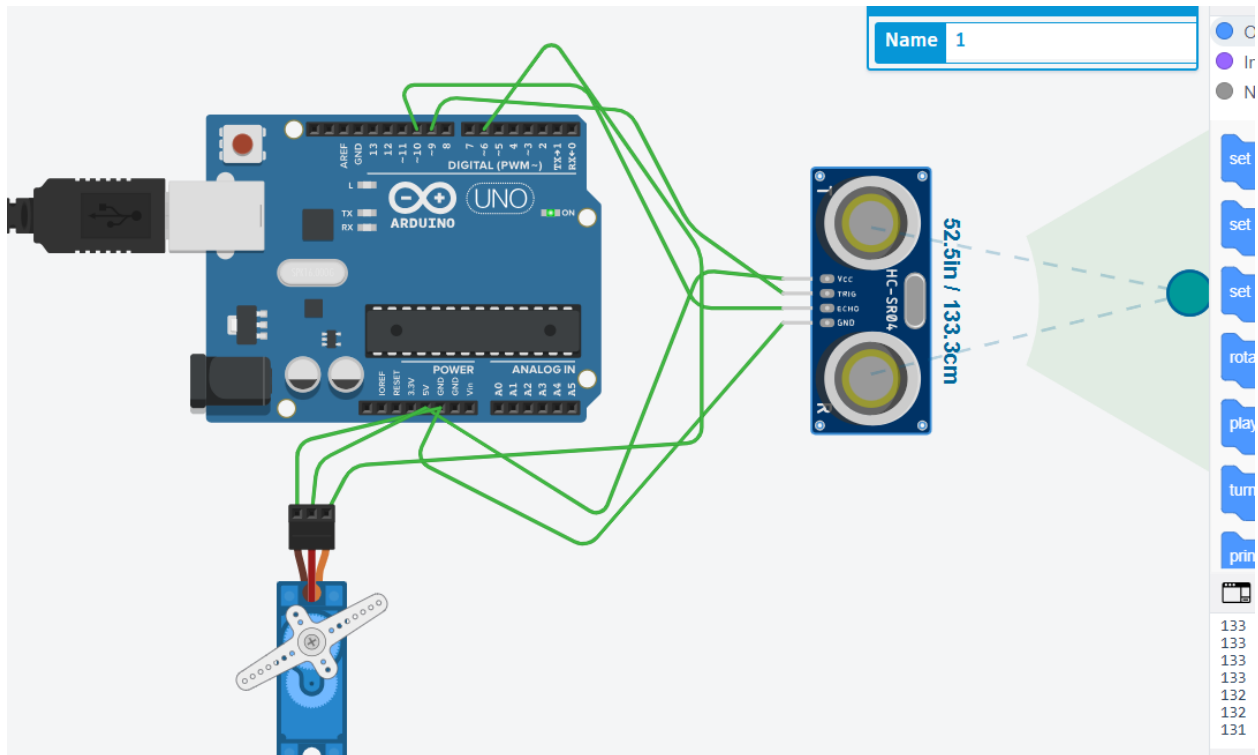
Print ncit:



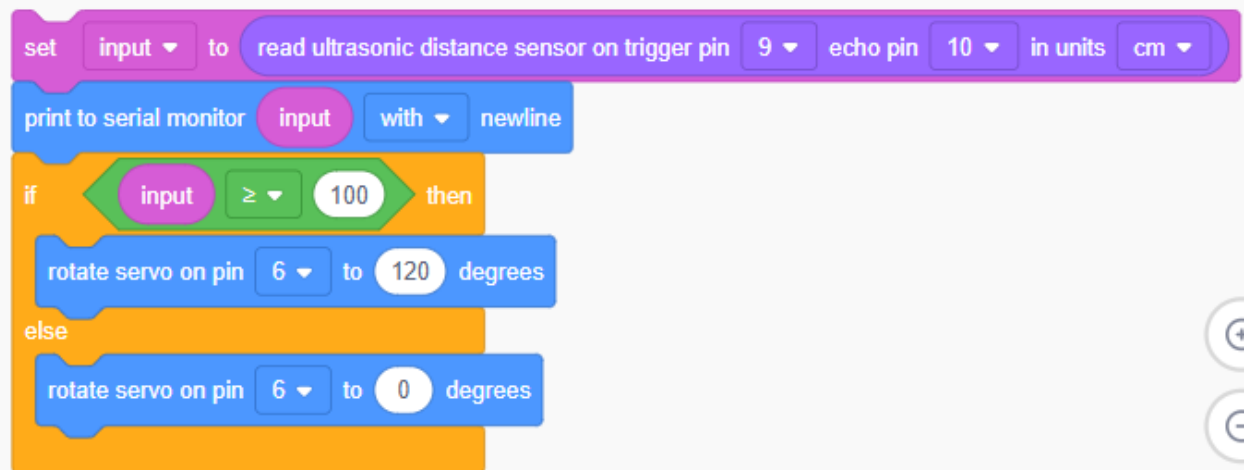
Code:



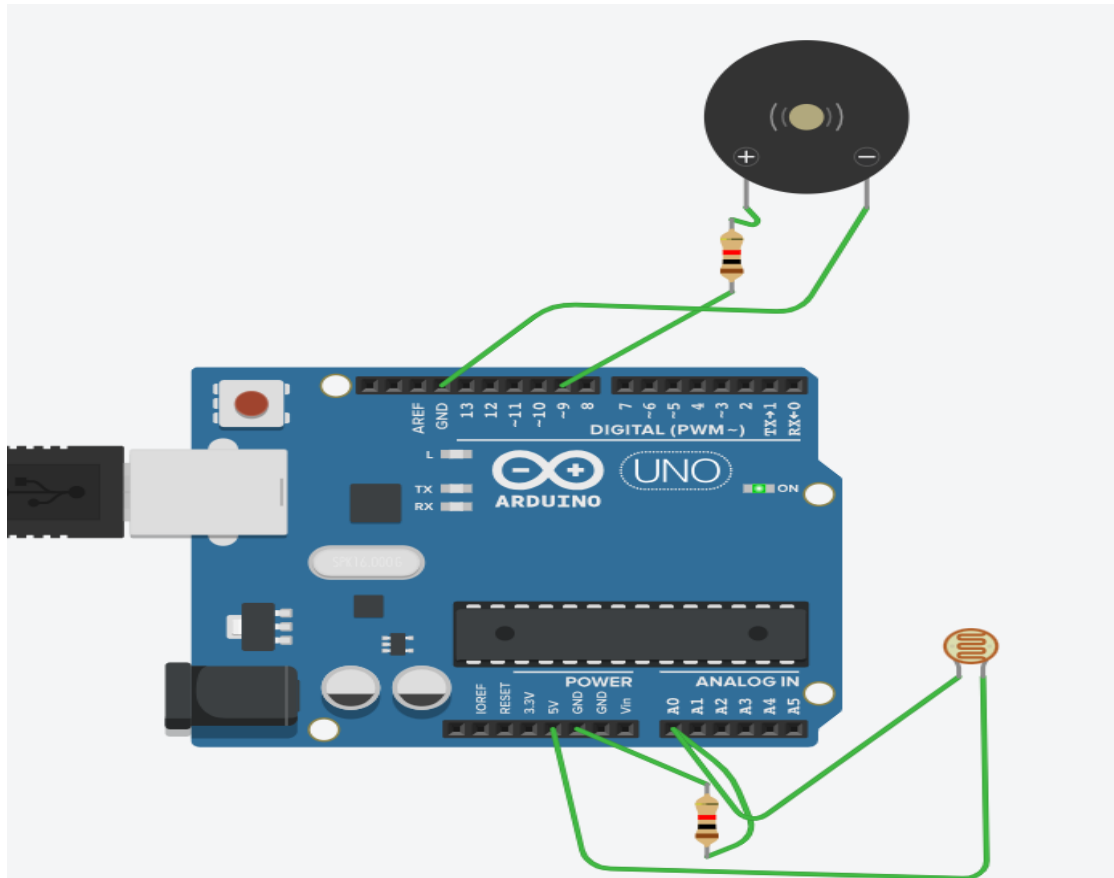
10. Ultrasonic sensor with servo motor:



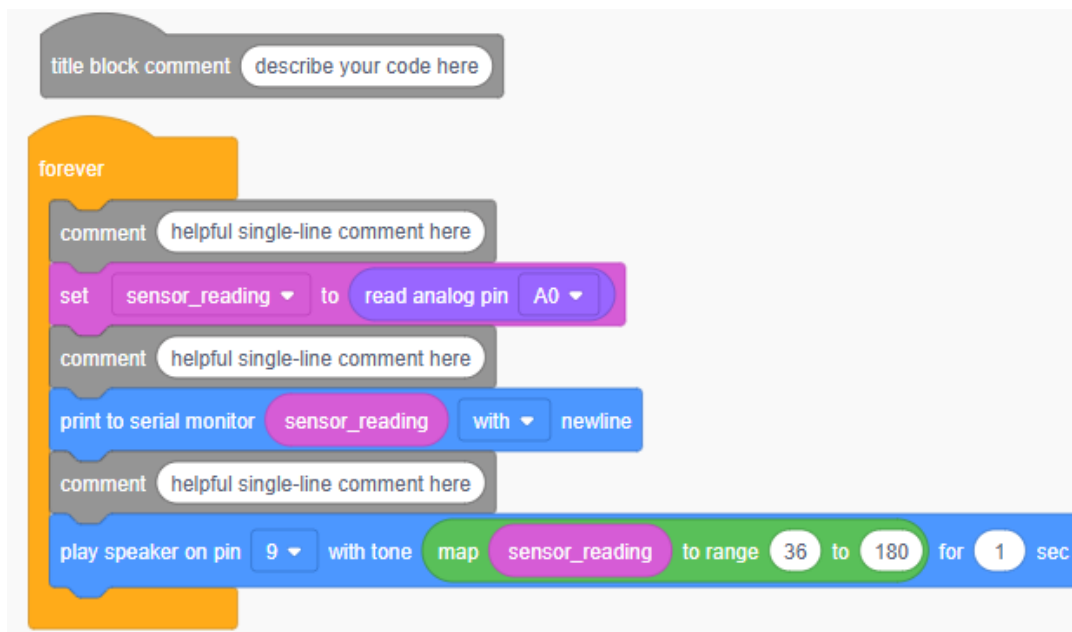
Code:



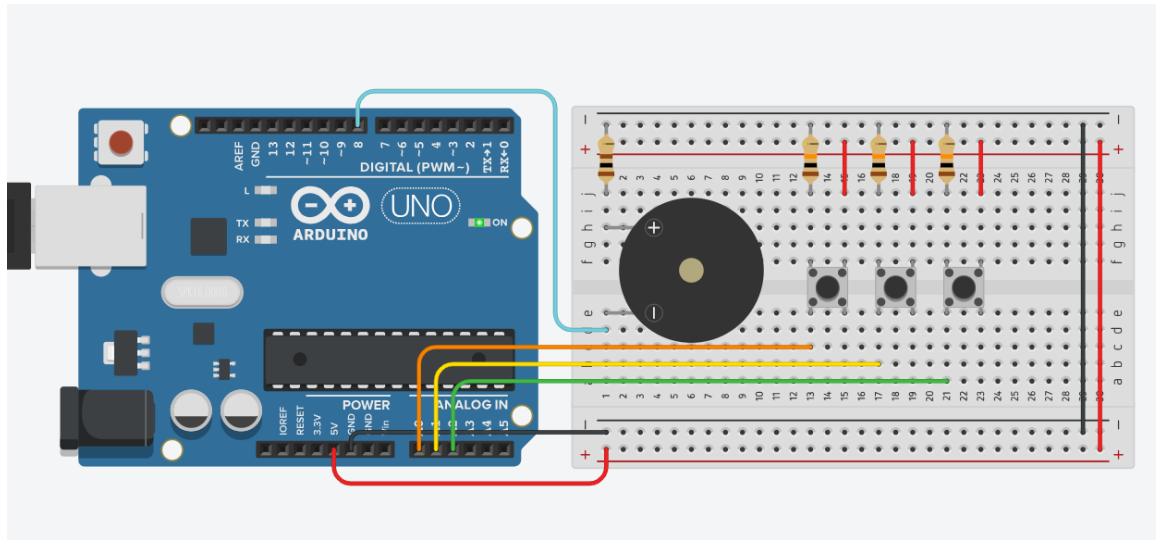
11. Tone Pitch follower:



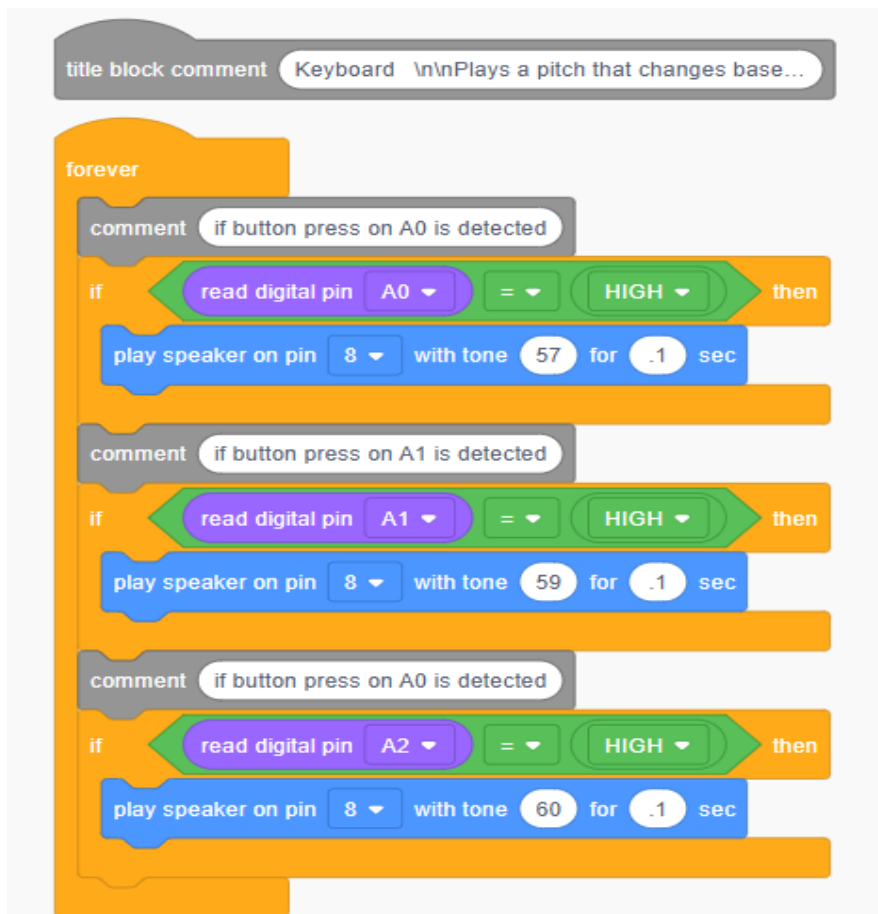
Code:



12. Tone keyboard:



Code:



Name 1

▼

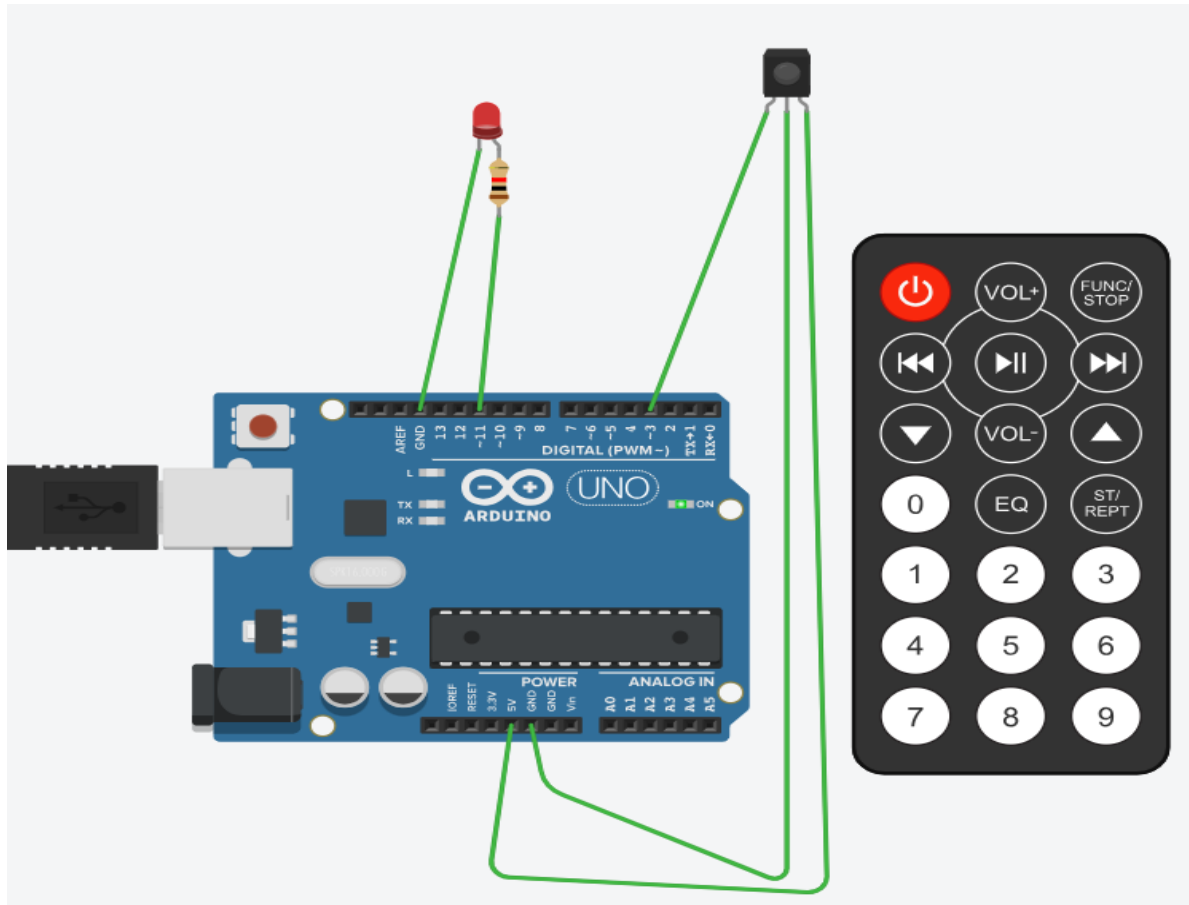


1 (Arduino Uno R3

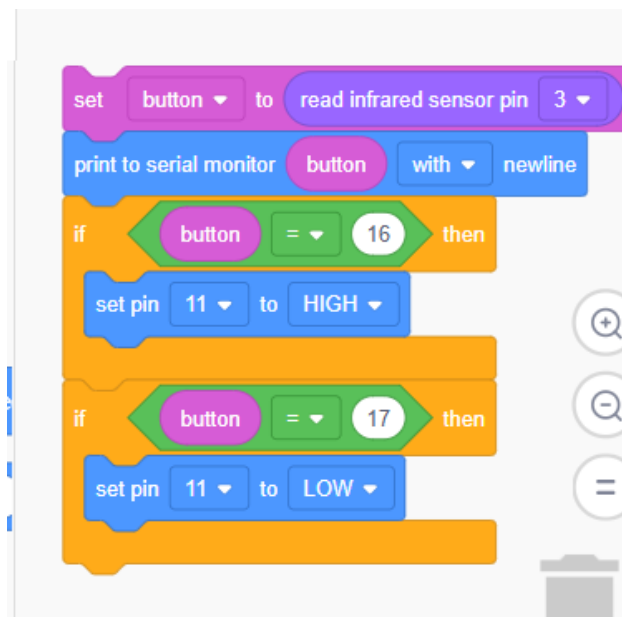
३३

[illegible]

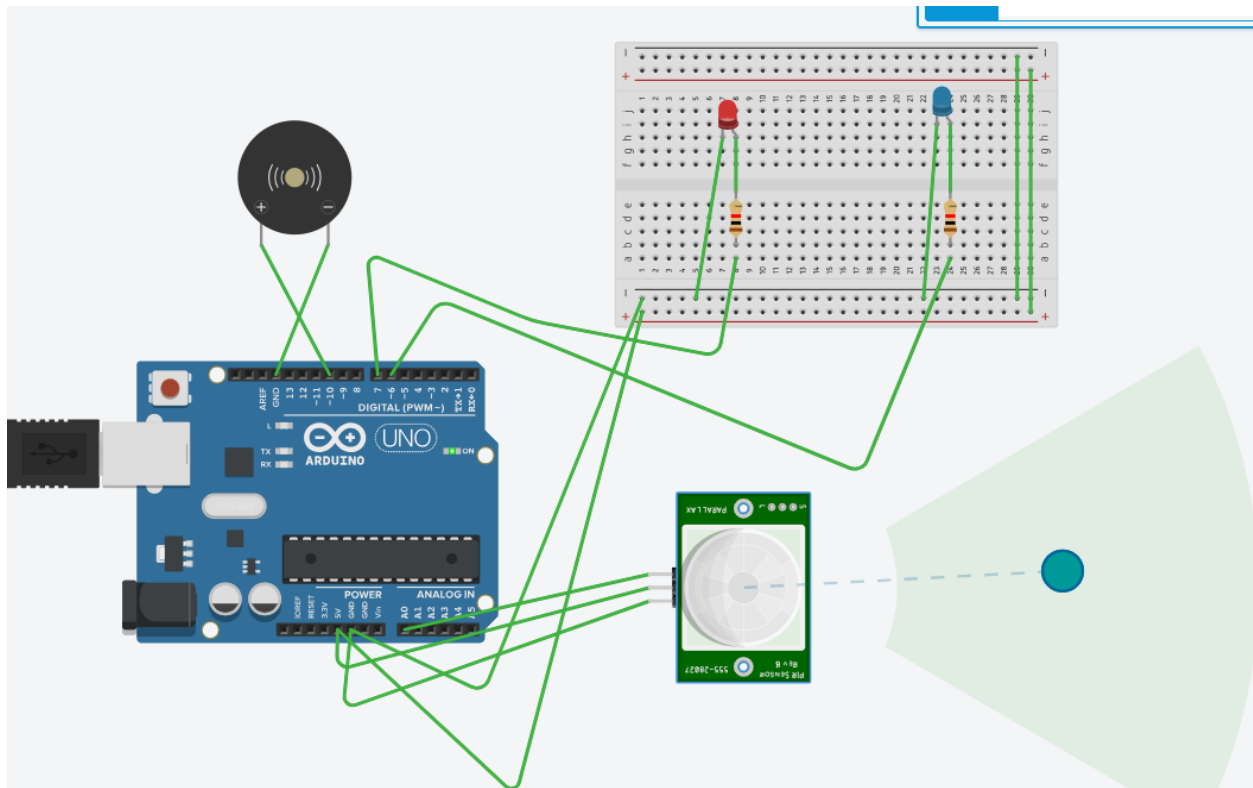
14. Controlling LEDs using an IR Remote:



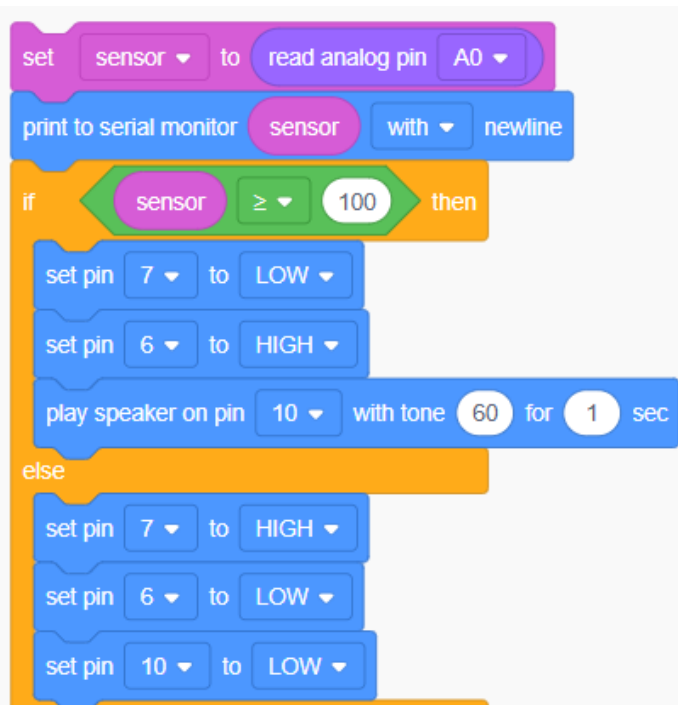
Code:



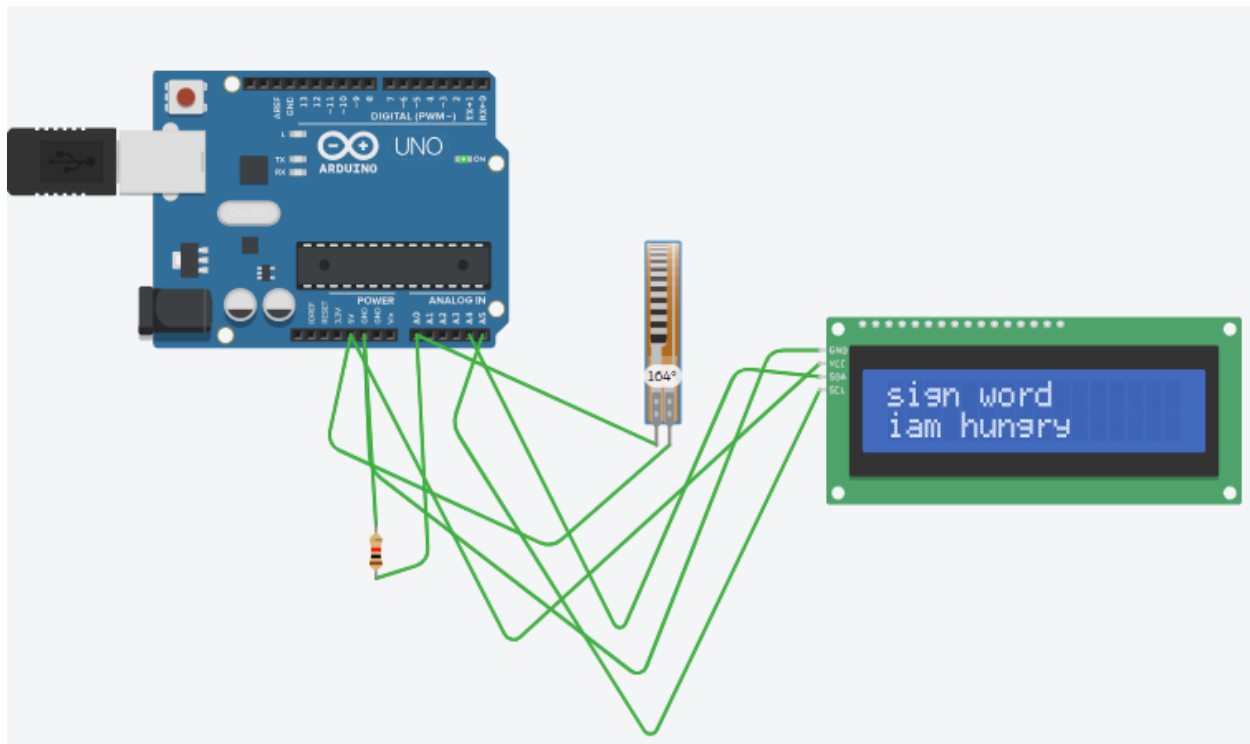
15. PIR motion sensor:



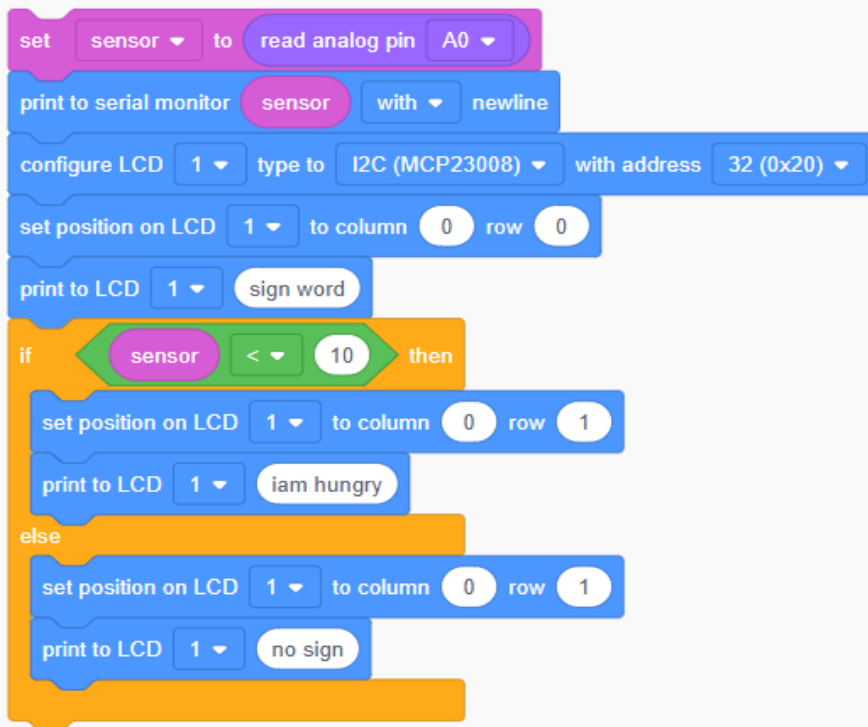
Code:



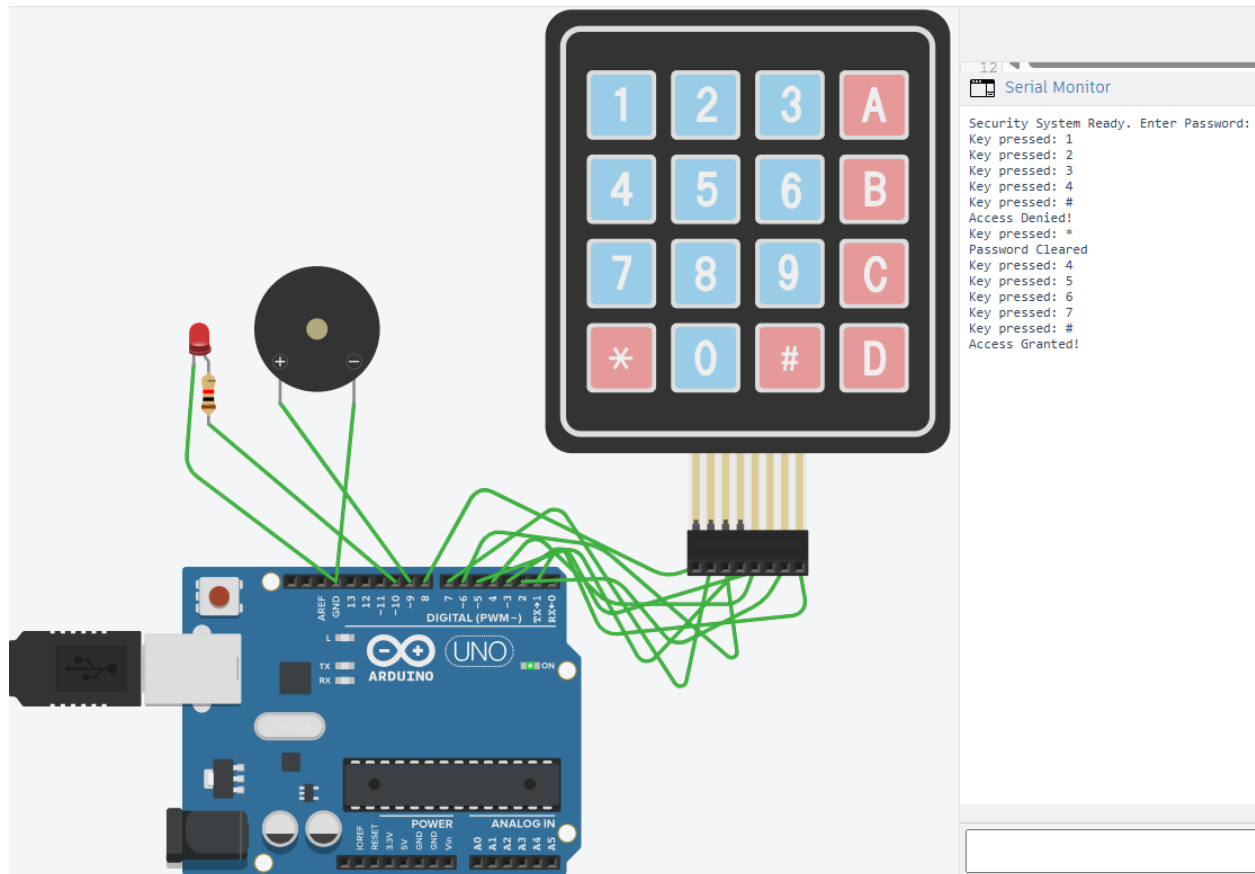
16. Flex sensor for sign language:



Code:



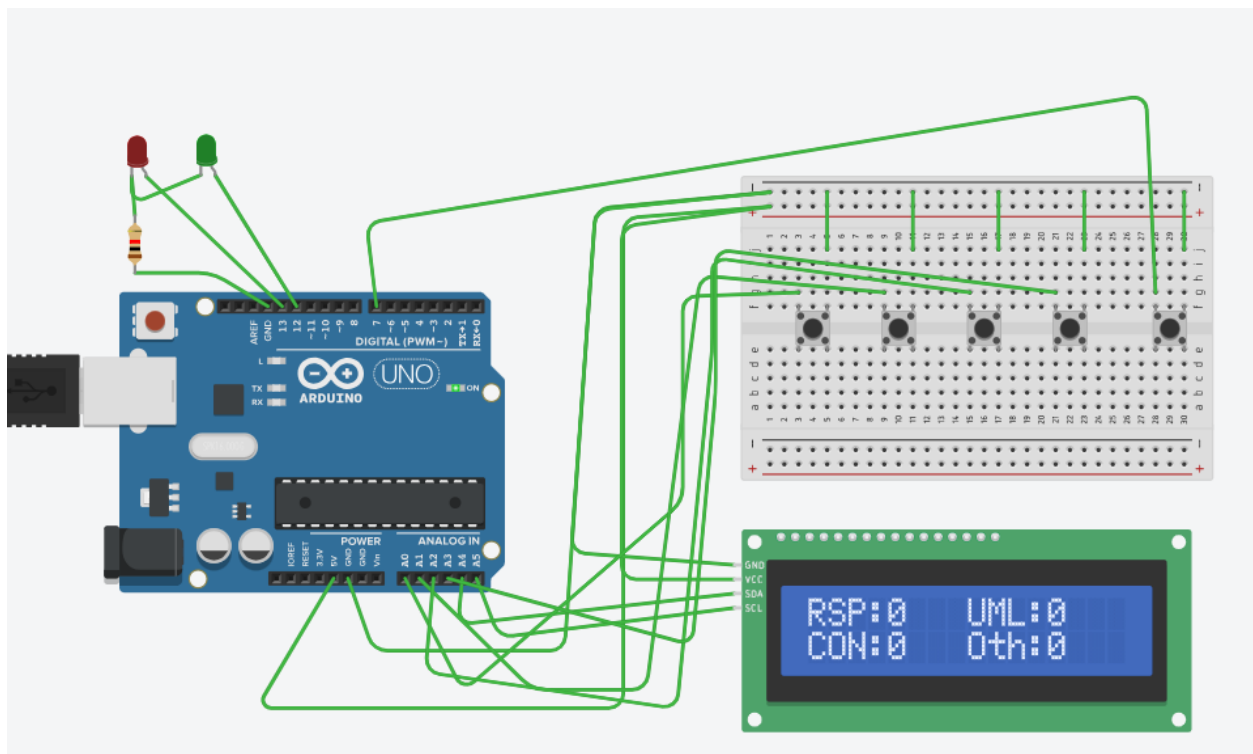
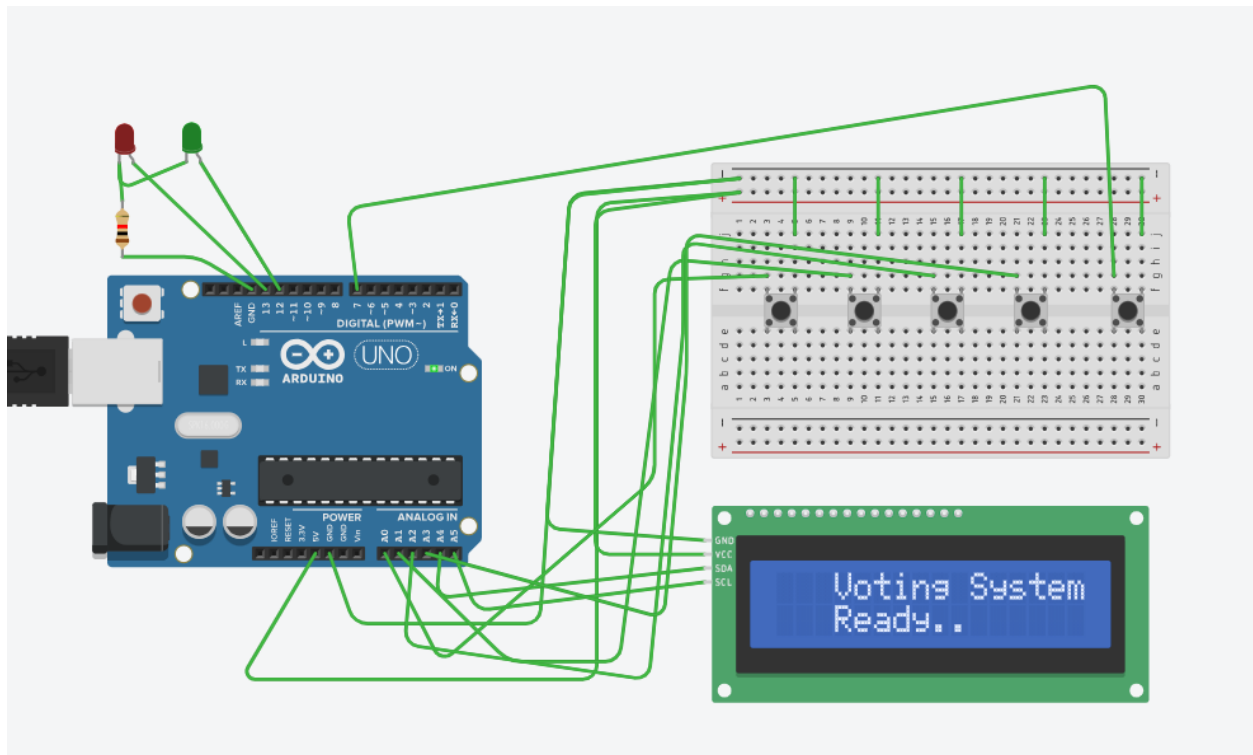
17. Password-Protected Security System project using Arduino, Keypad, Piezo Buzzer, and LED.

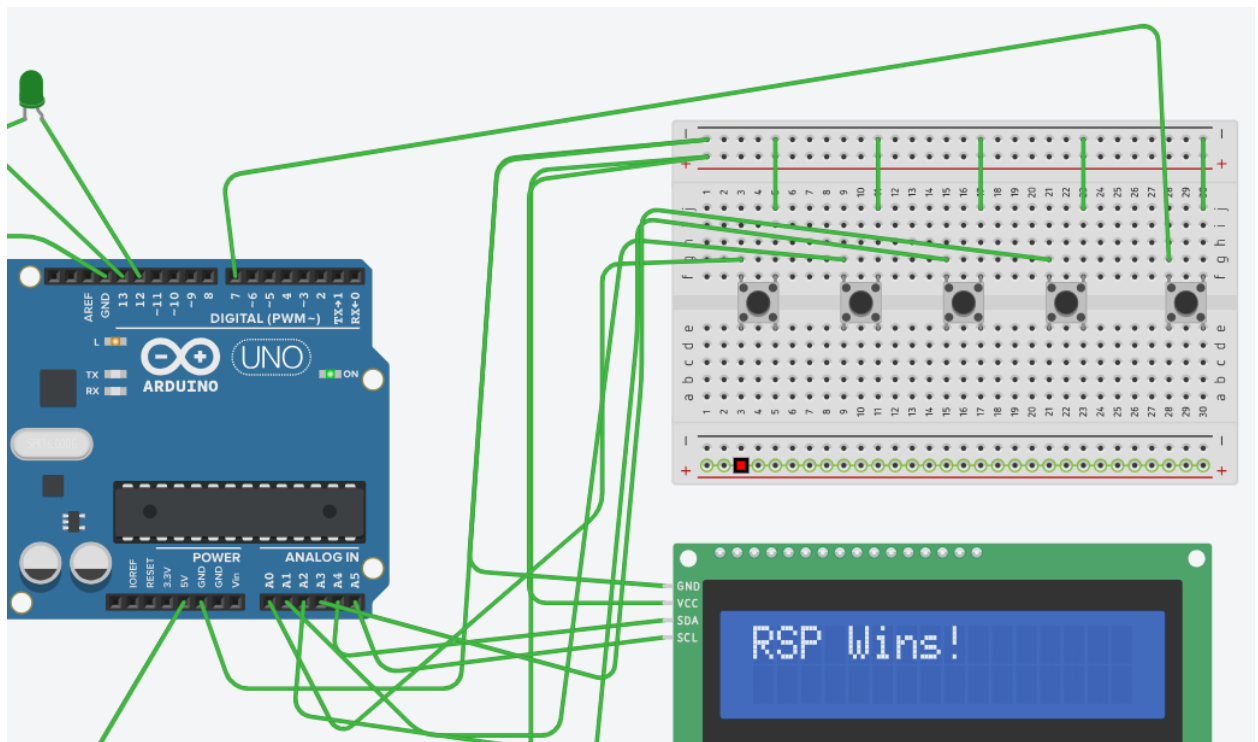
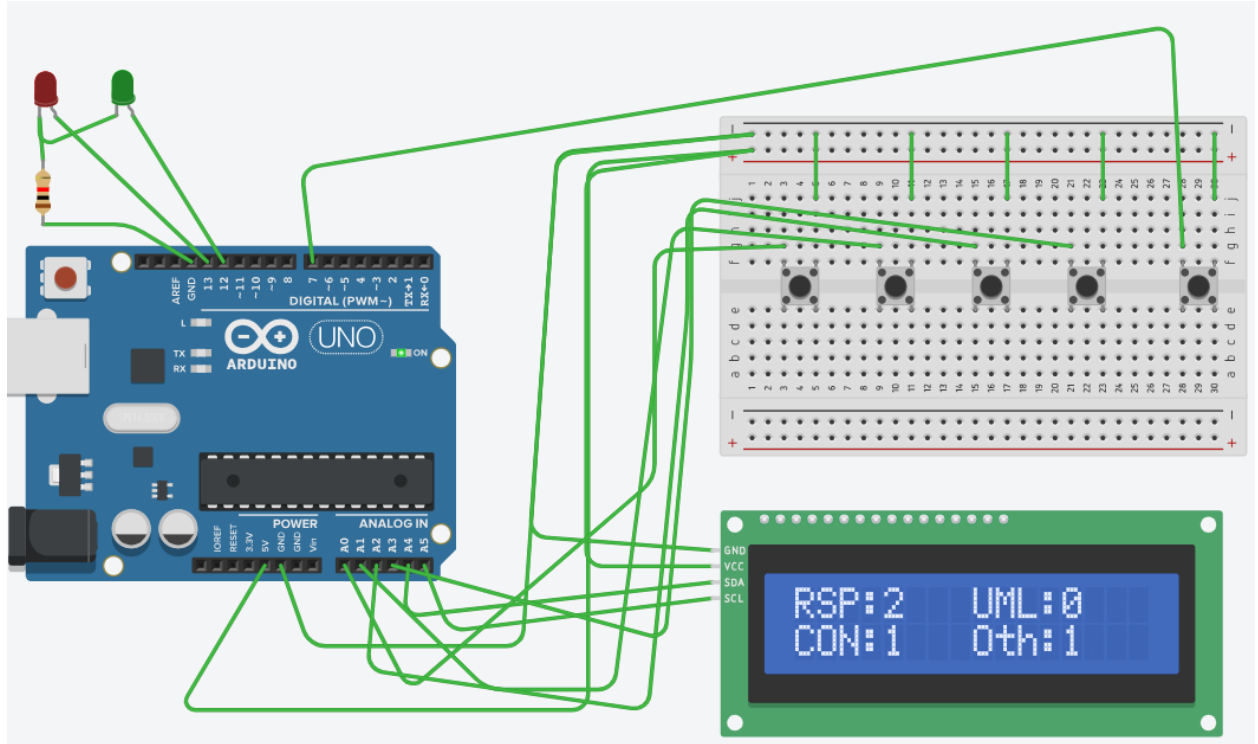


Code:

```
1 #include <Keypad.h>
2 const byte ROWS = 4;
3 const byte COLS = 4;
4 char keys[ROWS][COLS] = {
5     {'1','2','3','A'},
6     {'4','5','6','B'},
7     {'7','8','9','C'},
8     {'*','0','#','D'} };
9 byte rowPins[ROWS] = {8,7,6,5};
10 byte colPins[COLS] = {4,3,2,1};
11 Keypad keypad = Keypad(makeKeymap(keys), rowPins, colPins, ROWS,
12 const int ledPin = 10;
13 const int buzzerPin = 9;
14 String password = "4567";
15 String inputPassword = "";
16 void setup() {
17     pinMode(ledPin, OUTPUT);
18     pinMode(buzzerPin, OUTPUT);
19     Serial.begin(9600);
20     Serial.println("Security System Ready. Enter Password:");
21 }
22 void loop() {
23     char key = keypad.getKey();
24
25     if (key) {
26         Serial.print("Key pressed: ");
27         Serial.println(key);
28         if (key == '#') {
29
30             if (inputPassword == password) {
31                 Serial.println("Access Granted!");
32                 digitalWrite(ledPin, HIGH);
33                 noTone(buzzerPin);
34             } else {
35                 Serial.println("Access Denied!");
36                 digitalWrite(ledPin, LOW);
37                 tone(buzzerPin, 1000, 500);
38             }
39             inputPassword = "";
40         }
41         else if (key == '*') {
42             inputPassword = "";
43             Serial.println("Password Cleared");
44         }
45         else {
46             inputPassword += key;
47         }
48     }
49 }
```

18. Voting system with LCD:



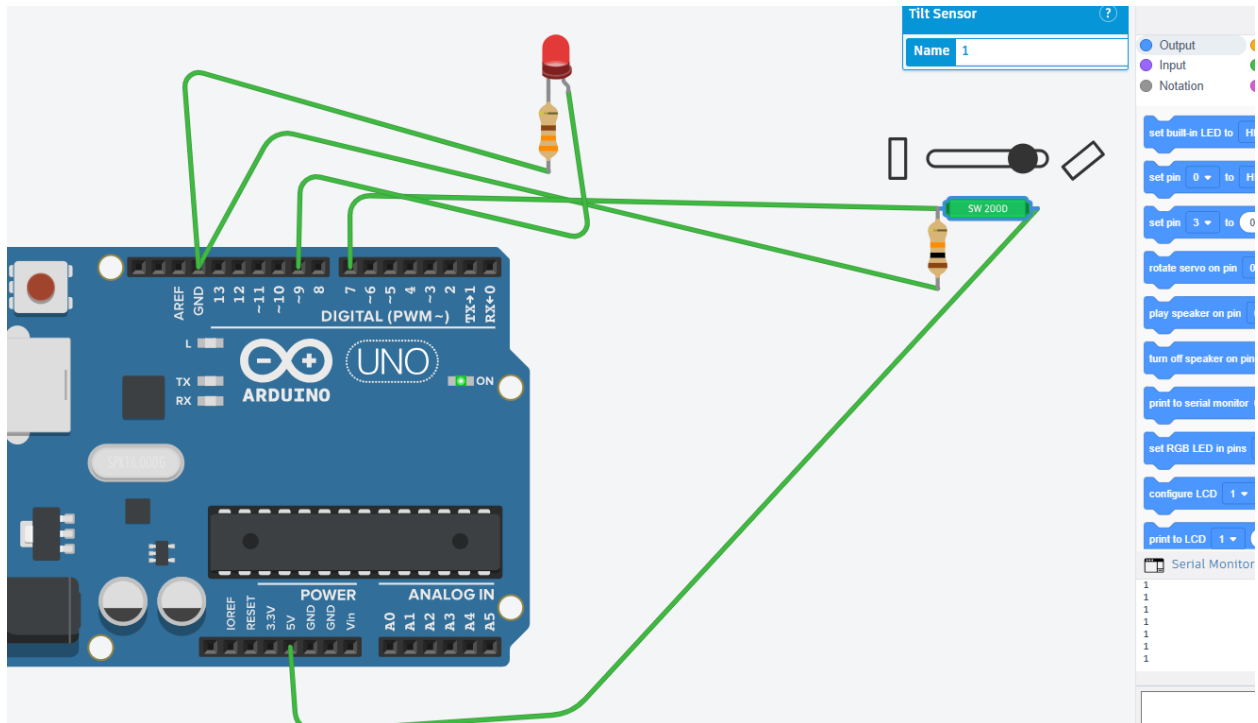


Code:

```
1  #include <Adafruit_LiquidCrystal.h>
2
3  Adafruit_LiquidCrystal lcd(0);
4  #define sw1 A0
5  #define sw2 A1
6  #define sw3 A2
7  #define sw4 A3
8  #define sw5 7    // Result button
9
10 int votes[4] = {0, 0, 0, 0};
11 void setup() {
12     pinMode(sw1, INPUT_PULLUP);
13     pinMode(sw2, INPUT_PULLUP);
14     pinMode(sw3, INPUT_PULLUP);
15     pinMode(sw4, INPUT_PULLUP);
16     pinMode(sw5, INPUT_PULLUP);
17     pinMode(12, OUTPUT); // Green LED
18     pinMode(13, OUTPUT); // Red LED
19     lcd.begin(16, 2);
20     lcd.setCursor(0,0);
21     lcd.print("    Voting System");|
22     lcd.setCursor(0,1);
23     lcd.print("    Ready..");
24     delay(2000);
25     lcd.clear();
26 }
27 void loop() {
28     lcd.setCursor(0,0); lcd.print("RSP:"); lcd.print(votes[0]);
29     lcd.setCursor(8,0); lcd.print("UML:"); lcd.print(votes[1]);
30     lcd.setCursor(0,1); lcd.print("CON:"); lcd.print(votes[2]);
31     lcd.setCursor(8,1); lcd.print("Oth:"); lcd.print(votes[3]);
32     if (digitalRead(sw1) == LOW) registerVote(0);
33     if (digitalRead(sw2) == LOW) registerVote(1);
34     if (digitalRead(sw3) == LOW) registerVote(2);
35     if (digitalRead(sw4) == LOW) registerVote(3);
36     if (digitalRead(sw5) == LOW) showResult();
37 }
38 void registerVote(int candidate) {
39     votes[candidate]++;
40     digitalWrite(12, HIGH); // Green LED ON
41     delay(100);
42     digitalWrite(12, LOW);
43     while (digitalRead(sw1+candidate) == LOW); // Wait until released
44 }
45
```

19. Light Intensity Measurement using LDR sensor: Circuit and code;

20. Vibration sensor:



Code:

